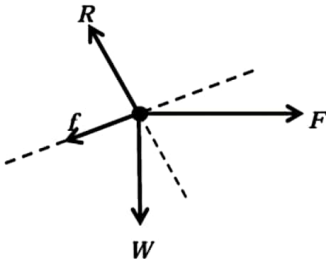
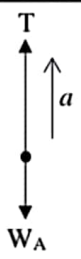


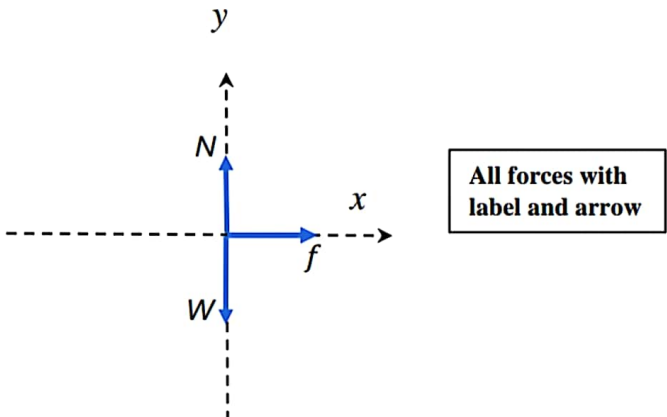
**SKEMA JAWAPAN DAN PEMARKAHAN UJIAN SELAR
SEMESTER I SESI 2024/2025**

NO.	ANSWER SCHEME															
1 (a)	$t = 8.2 \text{ minutes} = 492 \text{ s}$$v = 300000 \text{ km s}^{-1} = 3 \times 10^8 \text{ m s}^{-1}$$v = \frac{s}{t}$$s = 3 \times 10^8 \times 492$$s = 1.476 \times 10^{11} \text{ m}$ 1 / 6															
1 (b)	<table><tr><th>Vector</th><th>x-component</th><th>y-component</th></tr><tr><td>S_1</td><td>$35 \cos 40^\circ$ $= 26.811 \text{ km}$</td><td>$35 \sin 40^\circ$ $= 22.498 \text{ km}$</td></tr><tr><td>S_2</td><td>30 km</td><td>0 km</td></tr><tr><td>S_3</td><td>$-20 \cos 60^\circ$ $= -10 \text{ km}$</td><td>$20 \sin 60^\circ$ $= 17.321$</td></tr><tr><td>Resultant, R</td><td>46.81 km</td><td>39.82 km</td></tr></table> Magnitude , $S = \sqrt{S_x^2 + S_y^2}$ $S = \sqrt{46.81^2 + 39.82^2}$$S = 61.46 \text{ km}$ Direction, $\theta = \tan^{-1} \left(\frac{S_y}{S_x} \right)$ $= \tan^{-1} \frac{(39.82)}{(46.81)}$$= 40.4^\circ \text{ (above positive } x - \text{axis)}$	Vector	x-component	y-component	S_1	$35 \cos 40^\circ$ $= 26.811 \text{ km}$	$35 \sin 40^\circ$ $= 22.498 \text{ km}$	S_2	30 km	0 km	S_3	$-20 \cos 60^\circ$ $= -10 \text{ km}$	$20 \sin 60^\circ$ $= 17.321$	Resultant, R	46.81 km	39.82 km
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TC																
2 (a)	$u = 91 \text{ km h}^{-1} = 25.28 \text{ m s}^{-1}$$a = -6.40 \text{ m s}^{-2} \text{(negative is a must)}$$v^2 = u^2 + 2as$$0^2 = 25.28^2 + 2(-6.40)s$$s = 49.93 \text{ m}$															

2 (b)	(i) Displacement, $s = \text{area under the graph}$ $= \left(\frac{1}{2} \times (12 + 8) \times 18\right) + \left(\frac{1}{2} \times 4 \times 18\right) + \left(\frac{1}{2} \times 4 \times (-18)\right)$ $= 180 \text{ m}$ (ii) $a = \frac{v_f - v_i}{t_f - t_i}$ $a = \frac{0 - 18}{16 - 12}$ $a = -4.5 \text{ m s}^{-2}$	2 / 6 —
2 (c)	(i) $s = ut - \frac{1}{2}gt^2$ $0 = u(5) - \frac{1}{2}(9.81)(5^2) \quad (s = 0)$ $u = 24.53 \text{ m s}^{-1}$ (ii) $v^2 = u^2 + 2as$ $0^2 = 24.53^2 - 2(9.81)s \quad (\text{At maximum height, } v = 0)$ $s = 30.67 \text{ m}$	
TOTAL		
3 (a)	$\vec{J} = \text{Area under the graph}$ $= \frac{1}{2}[(4 + 7) \times 10^{-3}]8 \times 10^3$ $= 44 \text{ N s}$	
3 (b)	$\sum P_{ix} = \sum P_{fx}$ $m_A U_{Ax} + m_B U_{Bx} = (m_A + m_B)v_x$ $(m)(55 \sin 20^\circ) + m(55) = (m + m)v_x$ $v_x = 36.91 \text{ m s}^{-1}$ $\sum P_{iy} = \sum P_{fy}$ $m_A U_{Ay} + m_B U_{By} = (m_A + m_B)v_y$ $(m)(55 \cos 20^\circ) + m(0) = (m + m)v_y$ $v_y = 25.84 \text{ m s}^{-1}$ $\text{Magnitude, } v = \sqrt{(36.91)^2 + (25.84)^2}$ $= 45.06 \text{ m s}^{-1}$	

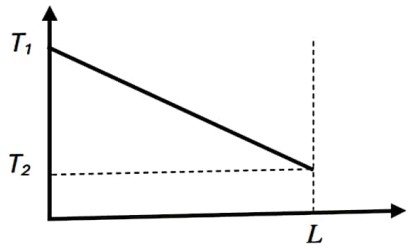
	$\text{Direction, } \alpha = \tan^{-1} \left(\frac{v_y}{v_x} \right)$ $= \frac{(25.84)}{(36.91)}$ $= 35^\circ$	C
TOTAL		
4 (a)	(i)  All 4 correct – D3 Any 3 correct – D2 Any 2 correct – D1 Others – 0 (ii) $\sum F_y = 0$ $N - mg \cos \theta - F \sin \theta = 0$ $N = mg \cos \theta + F \sin \theta$ $N = (2)(9.81) \cos 30^\circ + 50 \sin 30^\circ = 41.99 \text{ N}$ $\sum F_x = ma$ $F \cos \theta - mg \sin \theta - f = ma$ $F \cos \theta - mg \sin \theta - \mu N = ma$ $50 \cos 30^\circ - (2)(9.81) \sin 30^\circ - (0.25)(41.99) = 2a$ $a = 11.497 \text{ m s}^{-2}$	
4 (b)	<u>FBD Object A</u>  <u>FBD Object B</u>  $\sum F_y = m_A a$ $T - W_A = m_A a$ $T - (0.25)(9.81) = 0.25a$ $T - 2.45 = 0.25a$ $\sum F_y = m_B a$	

PHYSICS 1

	$W_B - T = m_B a$ $(0.5)(9.81) - T = 0.5a$ $4.91 - T = 0.5a$ $T = 3.27 \text{ N}$	
TOTAL		
5 (a)	(i) $v = r\omega$ $0.75 = (1.2)\omega$ $\omega = 0.63 \text{ rads}^{-1}$	
5 (a)	(ii) $\omega = \frac{\theta}{t}$ $\theta = \omega t$ $= (0.63)(120)$ $= 75.6 \text{ rad}$ $= \frac{75.6}{2\pi}$ $= 12.05 \text{ rev}$	
5 (b)	(i) 	
5 (b)	(ii) $\Sigma F_x = ma_c$ $0.55 = 0.1 a_c$ $a_c = 5.5 \text{ m s}^{-2}$	
TOTAL		
6 (a)	(i) $\omega = \omega_0 + \alpha t$ $10\pi = 20\pi + \alpha(6.0)$ $\alpha = -5.24 \text{ rads}^{-2}$	
6 (a)	(ii) $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$ $\theta = (20\pi)(6.0) + \frac{1}{2}(-5.24)(6.0)^2 \text{ (ecf for } \alpha)$ $\theta = 282.76 \text{ rad}$ $\theta = \frac{282.76}{2\pi}$ $\theta = 45 \text{ rotations}$ OR	

PHYSICS 1

	$\omega^2 = \omega_0^2 + 2\alpha\theta$ $(10\pi)^2 = (20\pi)^2 + 2(-5.24)\theta \quad (\text{ecf for } \alpha)$ $\theta = 282.76 \text{ rad}$ $\theta = \frac{282.76}{2\pi}$ $\theta = 45 \text{ rotations}$
6 (a)	(iii) $\omega = \omega_0 + \alpha t$ $0 = 10\pi + (-5.24)t$ $t = 6.0 \text{ s}$ OR $\omega = \omega_0 + \alpha t$ $0 = 20\pi + (-5.24)t$ $t = 12 \text{ s}$ Extra time needed: $t_{\text{extra}} = 12 - 6 = 6 \text{ s}$
6 (b)	(i) Pivot at A: $\Sigma\tau = F_1 r \sin \theta - W_L r \sin \theta + F_2 r \sin \theta$ $= (15)(1.4) \sin 90^\circ - (50)(3.5)(\sin 65^\circ) + (22)(6)(\sin 143^\circ)$ $= -58.16 \text{ N m}$
6 (b)	(ii) Pivot at B: $\Sigma\tau = F_1 r \sin \theta - W_P r \sin \theta + W_L r \sin \theta$ $= (15)(4.6) \sin 90^\circ - (2)(9.81)(3)(\sin 65^\circ) + (50)(2.5)(\sin 65^\circ)$ $= 97.63 \text{ N m}$
TOTAL	
7 (a)	(i) $\frac{Q}{t} = -kA \frac{\Delta T}{L}$ $= -(0.84)(0.323) \frac{(30 - 48)}{0.005}$ $= 9.68 \times 10^2 \text{ J s}^{-1}$ (ii) $\frac{Q}{t} = 9.68 \times 10^2$ $Q = (9.68 \times 10^2)(60)$ $= 5.81 \text{ J}$

	(iii) Temperature, T (K or $^{\circ}\text{C}$)  Label axes – 1 m Shape – 1 m
7 (b)	(i) $v_{rms} = \sqrt{\frac{3RT}{M}}$ $= \sqrt{\frac{3(8.314)(273.15+2)}{32 \times 10^{-3}}}$ $= 463.1 \text{ m s}^{-1}$ Molar mass = $32 \times 10^{-3} \text{ kg mol}^{-1}$ (ii) $v_{rms} = \sqrt{\frac{3kT}{m}}$ $600 = \sqrt{\frac{3(1.38 \times 10^{-23})(273.15+25)}{m}}$ $m = 3.43 \times 10^{-26} \text{ kg}$
7 (c)	(i) $Q = \Delta U + W$ $\Delta U = Q - W = 1000 - 20$ $\Delta U = 980 \text{ J}$ (ii) $\Delta U = 0 \text{ J}$ $Q = 0 + (4.0 \times 10^3)$ $Q = 4.0 \times 10^3 \text{ J}$ $Q = +\text{ve}$ and $W = +\text{ve}$
TOTAL	
GRAND TOTAL	

